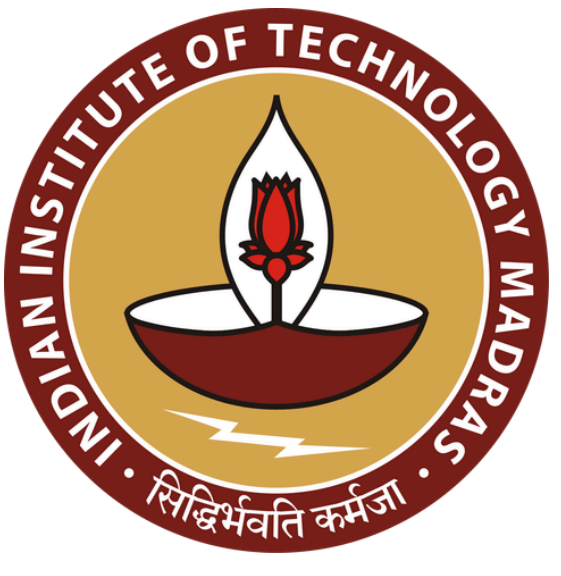




Dynamics of hierarchical navigation strategies in complex domains

FedEx SMART

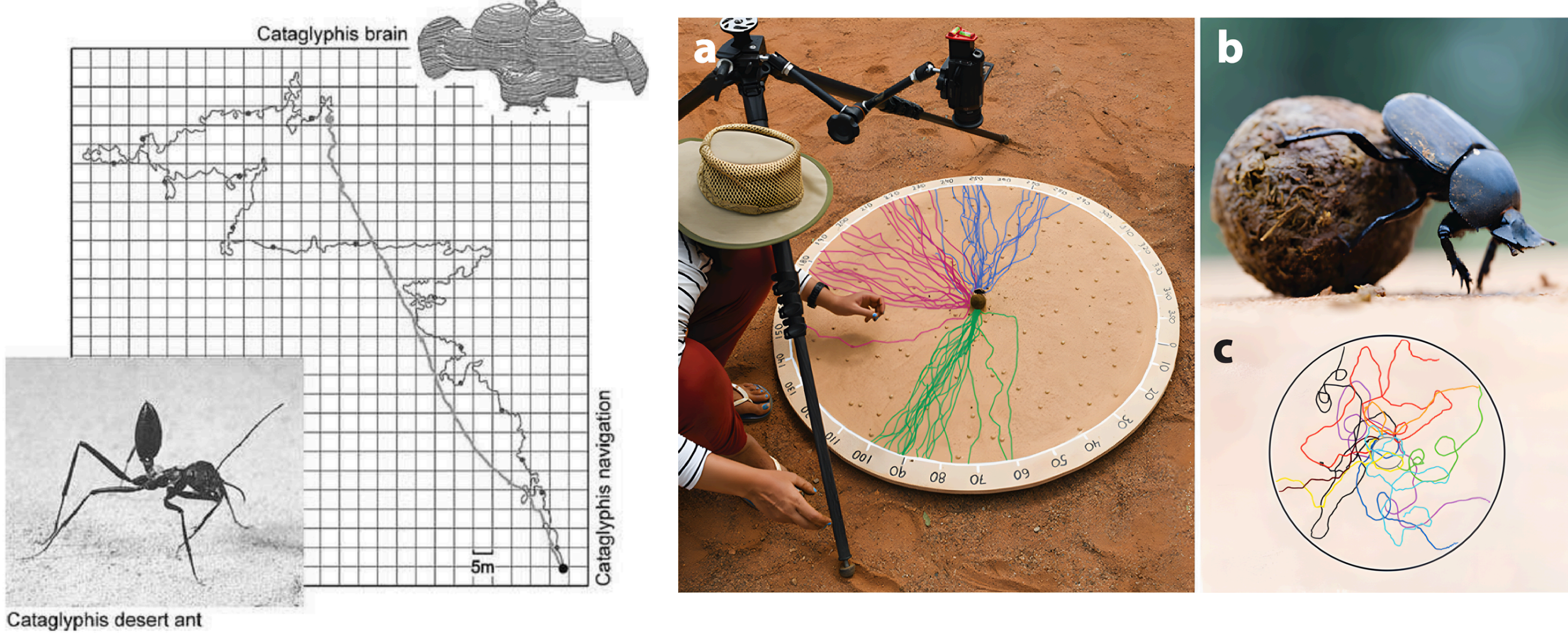
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INTERFERENCE lab



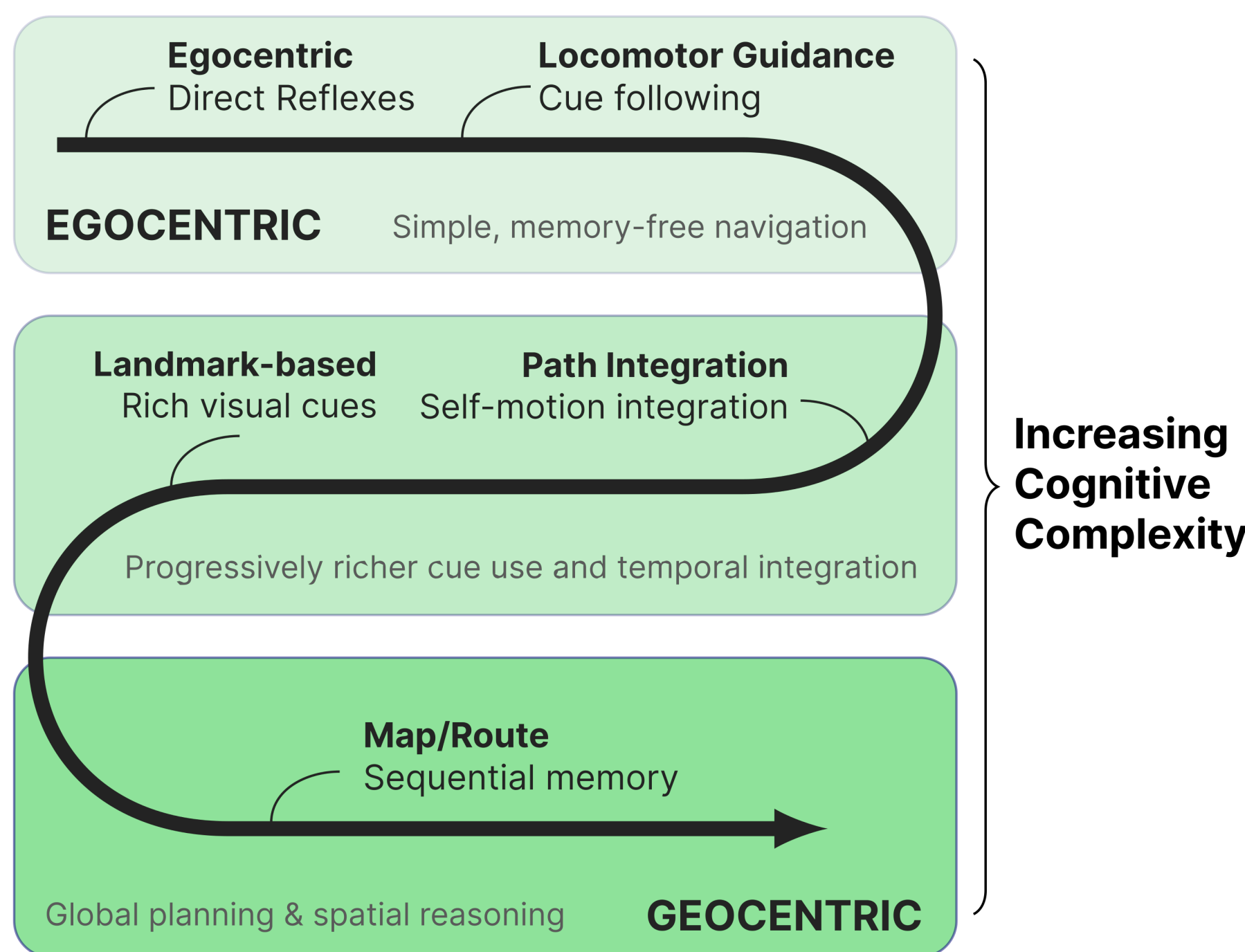
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Motivation

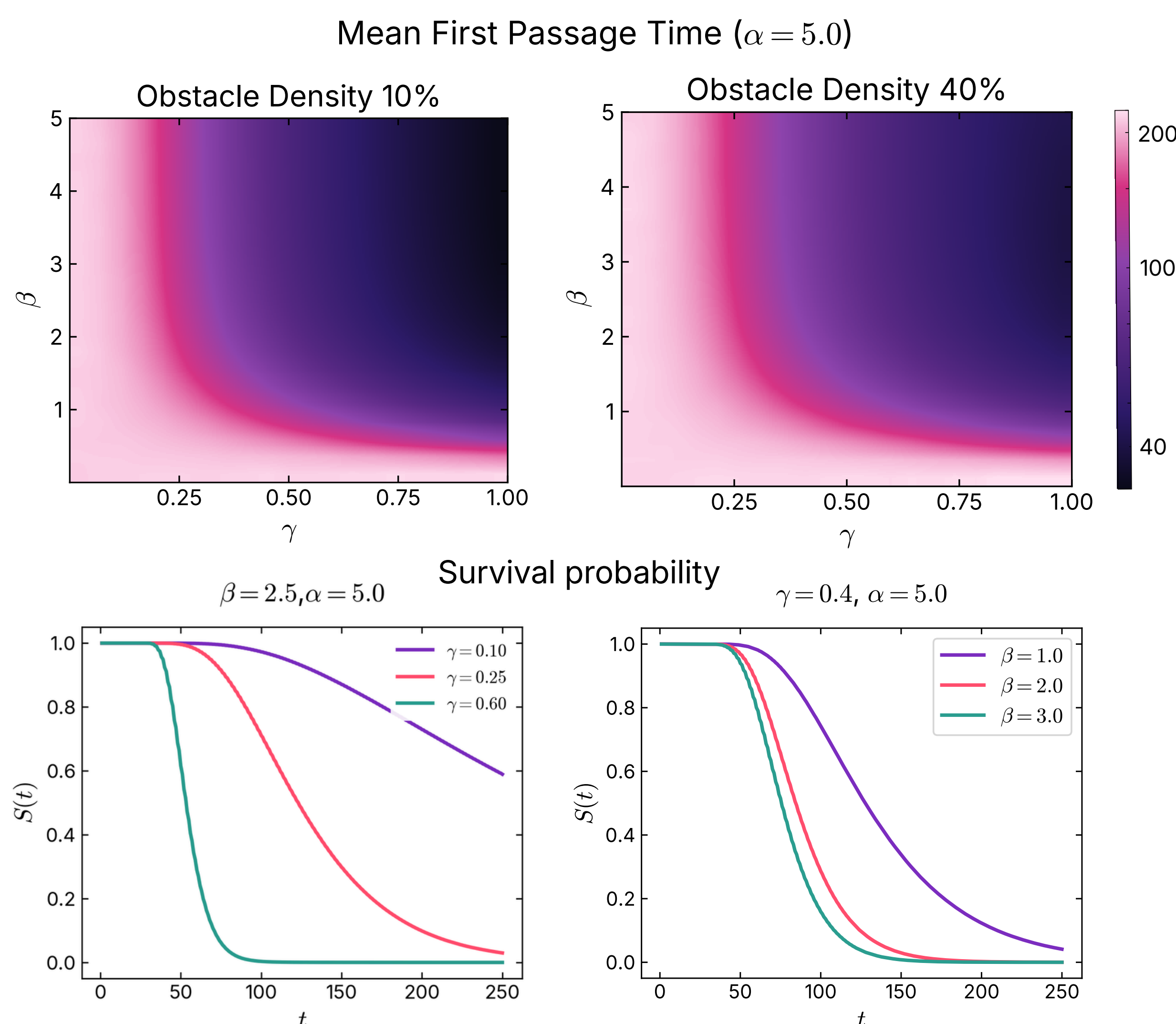


Insects such as desert ants and dung beetles use geocentric celestial cues to maintain straight, goal-directed navigation across featureless terrain. When these global cues degrade or uncertainty increases, they switch to egocentric strategies such as local search loops and landmark-based corrections. This adaptive switch integrates stable global heading with local feedback, ensuring robust navigation and precise target localization.

Navigation evolves from direct egocentric reflexes with no memory to progressively richer cue use, sequential memory, and self-motion integration, leading to a geocentric map enabling global planning. How do agents make decisions on which strategy to choose depending on their sensory information? How did these strategies evolve and do they optimise any cost function?



Preliminary Results



Conclusions & Open questions

- Policy framework naturally fits into Master equation paradigm
- Agents can weight between policy depending on local context
- What is the optimal γ as a function of obstacle density?
- What are the connections with classical biased random walk models?
- Dynamics of generalisation: What is the performance of best performing low disorder density policy in high density?
- Can generalisation be predicted from the statistical properties of first-passage time distributions under environment perturbations?
- What is the reward that produces sum of softmax policies?

Model

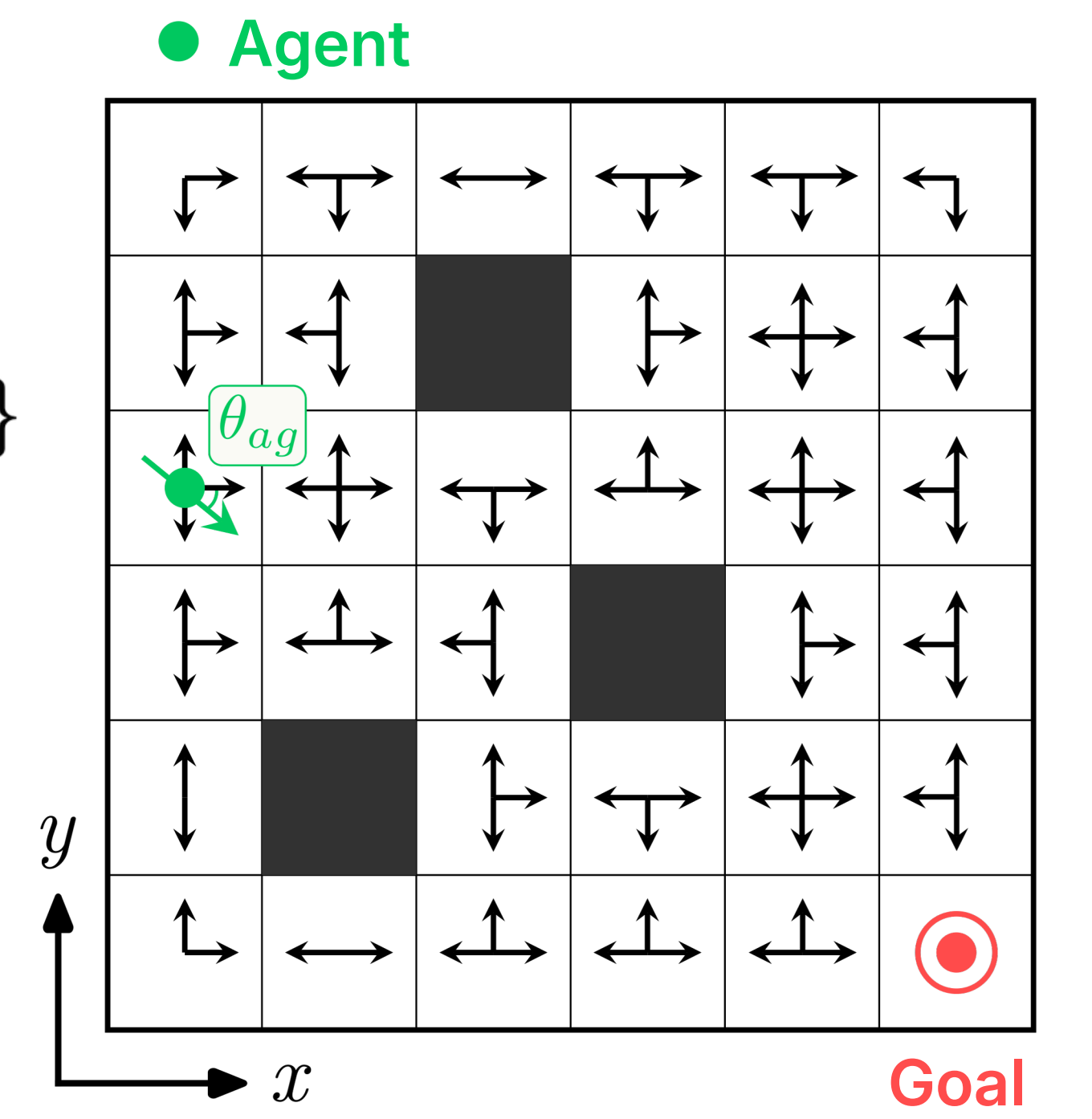
Gridworld

$$\text{State, } s = \{x, y\}$$

$$\text{Actions, } a \in \{+\hat{x}, -\hat{x}, +\hat{y}, -\hat{y}\}$$

$$\text{Homing vector orientation, } \theta_{ag} \in [0, 2\pi)$$

$$\text{Occupancy, } O_a \in \{0, 1\}$$



Geocentric policy

$$\pi_G(a|s) = \frac{\exp(\beta\phi_a)}{\sum_{a'} \exp(\beta\phi_{a'})}$$

Egocentric policy

$$\pi_E(a|s) = \frac{\exp(-\alpha O_a)}{\sum_{a'} \exp(-\alpha O_{a'})}$$

$$\pi(a|s) = \gamma \pi_G(a|s) + (1 - \gamma) \pi_E(a|s) \text{ where } \gamma \in [0, 1]$$

Master equation

$$\frac{\partial P(\bar{r}, t)}{\partial t} = \sum_{\bar{a}} [\pi(\bar{a} | \bar{r} - \bar{a}) P(\bar{r} - \bar{a}, t) - \pi(\bar{a} | \bar{r}) P(\bar{r}, t)]$$

Continuum limit

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial x} [V_x(\bar{r}) P] - \frac{\partial}{\partial y} [V_y(\bar{r}) P] + D_{\parallel} \frac{\partial^2 P}{\partial x^2} + D_{\perp} \frac{\partial^2 P}{\partial y^2}.$$

$$\begin{aligned} V_x(\bar{r}) &= v_0 - \frac{(1-\gamma)\alpha\epsilon^2}{2} \partial_x O(\bar{r}), & D_{\perp} &= \frac{\gamma\epsilon^2}{2(1+\cosh\beta)} + \frac{(1-\gamma)\epsilon^2}{2} \\ V_y(\bar{r}) &= \kappa\theta(\bar{r}) - \frac{(1-\gamma)\alpha\epsilon^2}{2} \partial_y O(\bar{r}), & D_{\parallel} &= \frac{\gamma\epsilon^2 \cosh\beta}{2(1+\cosh\beta)} + \frac{(1-\gamma)\epsilon^2}{2} \\ v_0 &= \gamma\epsilon \tanh \frac{\beta}{2} & \kappa &= \frac{\gamma\epsilon\beta}{4 \cosh^2(\beta/2)} \end{aligned}$$

Homogeneous obstacle density

$$\beta \ll 1 \quad f_P(t) = \frac{L}{\epsilon\sqrt{2\pi}t^3} \exp\left(-\frac{(L-\gamma\epsilon t)^2}{2\epsilon^2 t}\right)$$

First passage time distribution

$$f_P(t) = -\frac{dS(t)}{dt} \quad \beta \gg 1 \quad f_P(t) = \frac{L}{\epsilon\sqrt{\pi(2-\gamma)t^3}} \exp\left(-\frac{(L-v_0 t)^2}{\epsilon^2(2-\gamma)t}\right)$$

Optimal transport

$$\text{SDE corresponding to Fokker-Planck equation: } d\mathbf{r} = \mathbf{V}(\mathbf{r}) dt + \sqrt{2\mathbf{D}} d\mathbf{W}$$

$$\text{What is the corresponding reward minimised by the SDE? } J = \mathbb{E} \left[\int_0^T \left(\frac{1}{2} |\mathbf{u}|^2 - R(\mathbf{r}, t) \right) dt \right]$$

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